Block practice · Vectors

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Getting from A to B: subtracting position vectors

Finding a route: vectors round a shape with midpoints and ratios

How long is a vector? Magnitudes and finding unknowns

Unit vectors: same direction, length one

Proving shapes and straight lines with vectors

Where lines cross: two routes and comparing coefficients

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Practice workbook · Getting from A to B: subtracting position vectors

Finding the vector joining two points, or the position vector of a point, by subtracting or adding position vectors given in i, j or in letter vectors such as a and b.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-01

1. Question 1 · part a · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part a only.

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1 Referred to a fixed origin O, the point A has position vector $(4\mathbf{i} + 3\mathbf{j})$ and the point B has position vector $(\mathbf{i} + 7\mathbf{j})$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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(Total for Question 1 is 4 marks)

1. Question 1 · part a · 2 marks · Working and answer

Working and answer

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2. Question 7 · part a · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 O, A, B and C are fixed points such that

$$\overrightarrow{OA} = 8\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OB} = 15\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OC} = 8\mathbf{i} + \mathbf{j}$$

(a) Find $\overrightarrow{BC}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector parallel to $\overrightarrow{BC}$

(2)

The point M is the midpoint of OA and the point N lies on OB such that $ON : NB = 1 : 2$

(c) Show that the points M, N and C are collinear.

(4)

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2. Question 7 · part a · 2 marks · Working and answer

Working and answer

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3. Question 1 · part a · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

1 The position vector of the point A is $(3\mathbf{i} - 2\mathbf{j})$, referred to a fixed origin O.

The point B is such that $\overrightarrow{AB} = (6\mathbf{i} + 8\mathbf{j})$

(a) Find the position vector of B as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find the magnitude of vector $\overrightarrow{AB}$

(1)

(c) Find a unit vector, in terms of $\mathbf{i}$ and $\mathbf{j}$, that is parallel to $\overrightarrow{AB}$

(2)

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3. Question 1 · part a · 2 marks · Working and answer

Working and answer

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4. Question 10 · part a.i · 0 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

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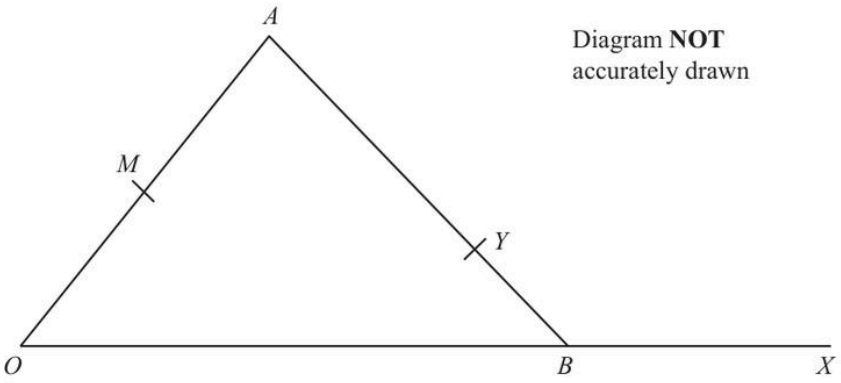


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = 2\mathbf{a}$, $\overrightarrow{OB} = 4\mathbf{b}$ and M is the midpoint of OA.

The point Y lies on AB such that $AY : YB = 3 : 1$

The point X lies on OB produced.

(a) Find as simplified expressions in terms of **a** and **b**

(i) $\overrightarrow{AB}$ (ii) $\overrightarrow{MY}$

(3)

The points M, Y and X are collinear.

(b) Find the ratio $OB : OX$

(5)

(c) Find the ratio of $(\text{Area } \Delta YBX) : (\text{Area } \Delta OAX)$

(3)

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4. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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5. Question 8 · part a.i · 0 marks

2024 NOV · Paper 02

Further practice · Attempt part a.i only.

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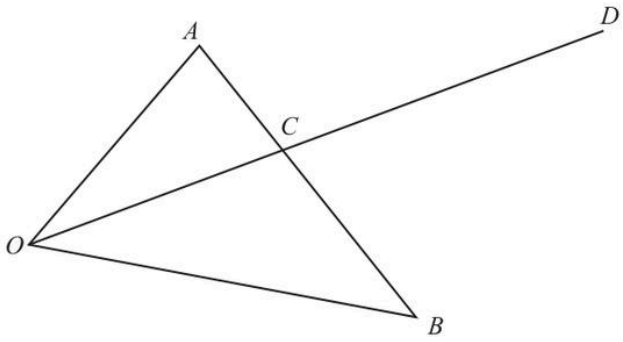


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$

(3)

(b) Find the area of triangle AOB

(4)

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of **a** and **b**

(5)

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P 7 3 5 8 7 A 0 2 0 3 2

5. Question 8 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 4 · part b · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = p\mathbf{i} + 2p\mathbf{j} \qquad \overrightarrow{OB} = 5\mathbf{i} + 9p\mathbf{j}$$

Given that $\overrightarrow{AB}$ is parallel to $(\mathbf{i} - 2\mathbf{j})$

(a) find the value of p

(6)

(b) Hence find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(c) Find a unit vector parallel to $\overrightarrow{OA}$

Give your answer in the form $\frac{\sqrt{a}}{5}(b\mathbf{i} + c\mathbf{j})$ where a, b and c are integers to be found.

(4)

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6. Question 4 · part b · 2 marks · Working and answer

Working and answer

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7. Question 9 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

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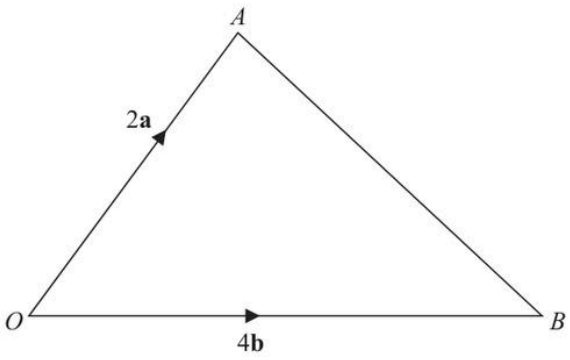


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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7. Question 9 · part a · 2 marks · Working and answer

Working and answer

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8. Question 8 · part a · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

8

Diagram **NOT** accurately drawn

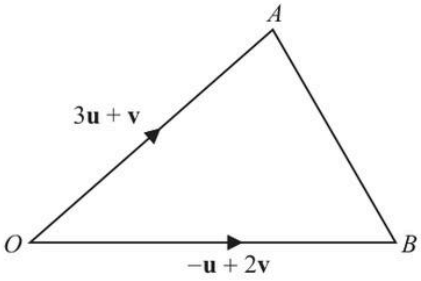


Figure 2

Figure 2 shows triangle OAB with

$$\overrightarrow{OA} = 3\mathbf{u} + \mathbf{v} \qquad \overrightarrow{OB} = -\mathbf{u} + 2\mathbf{v}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ **(2)**

Point C is such that AC is parallel to OB

Given that $\overrightarrow{OC} = \mu\mathbf{v}$

(b) find the value of μ **(4)**

The lines OC and AB intersect at point X

Point D lies on $\overrightarrow{OC}$ such that

area of triangle BOX : area of triangle $BXD = 2 : 3$

Using a vector method,

(c) find $\overrightarrow{BD}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ **(5)**

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8. Question 8 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Finding a route: vectors round a shape with midpoints and ratios

Building a vector in terms of given base vectors by travelling along known sides of a figure, including through midpoints and points that divide a line in a given ratio.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-04

1. Question 11 · part a.i · 3 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part a.i only.

11

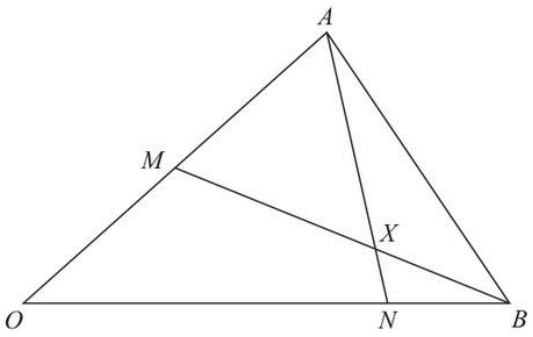


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

M is the midpoint of OA.

N is the point on OB such that $ON : NB = 3 : 1$

The lines AN and BM intersect at the point X.

(a) Find expressions, in terms of **a** and **b**, for

(i) $\overrightarrow{AN}$ (ii) $\overrightarrow{BM}$

(3)

(b) Using a vector method, find $AX : XN$

(7)

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P 7 1 8 1 7 A 0 3 0 3 2

1. Question 11 · part a.i · 3 marks · Working and answer

Working and answer

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2. Question 8 · part a · 1 mark

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8

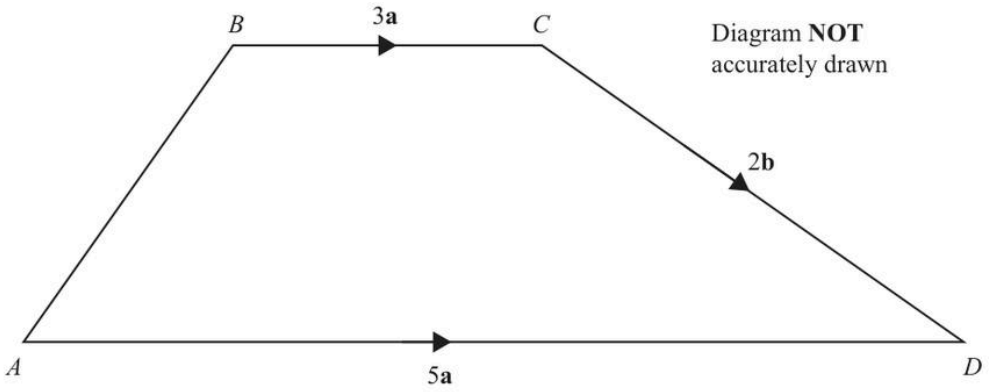


Figure 3

Figure 3 shows a trapezium $ABCD$

$\overrightarrow{BC} = 3\mathbf{a}$ $\overrightarrow{AD} = 5\mathbf{a}$ $\overrightarrow{CD} = 2\mathbf{b}$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

The diagonals BD and AC intersect at point X where $\overrightarrow{BX} = k\overrightarrow{BD}$

(b) Using a vector method, find the value of k

(5)

(c) Find the ratio of the area of triangle CXD : area of the trapezium $ABCD$

(4)

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2. Question 8 · part a · 1 mark · Working and answer

Working and answer

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3. Question 10 · part a · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

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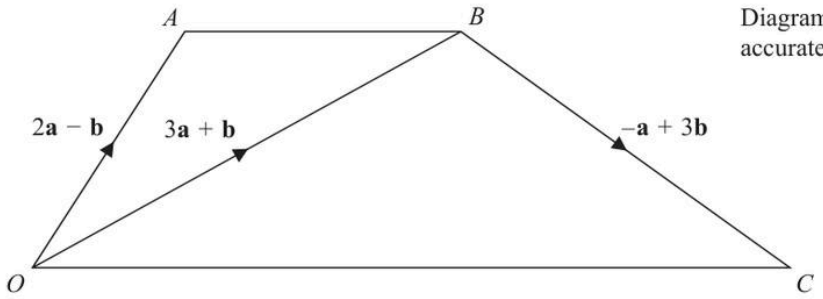


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows quadrilateral $OABC$ with

$$\overrightarrow{OA} = 2\mathbf{a} - \mathbf{b} \qquad \overrightarrow{OB} = 3\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = -\mathbf{a} + 3\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**.

(2)

(b) Prove that $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$

(2)

The diagonals, OB and AC, intersect at the point X.

(c) Using a vector method find the ratio $AX : XC$

(7)

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3. Question 10 · part a · 2 marks · Working and answer

Working and answer

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4. Question 10 · part a · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

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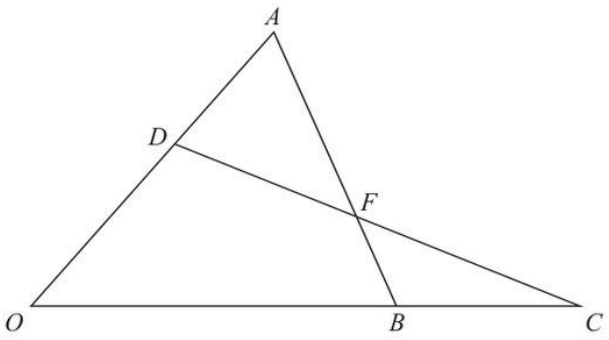


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle OAB and triangle OCD.

$$\overrightarrow{OA} = 5\mathbf{p} \quad \overrightarrow{AB} = 3\mathbf{q} \quad \overrightarrow{OC} = \frac{3}{2}\overrightarrow{OB} \quad \overrightarrow{OD} = \frac{3}{5}\overrightarrow{OA}$$

(a) Find $\overrightarrow{DC}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(3)

The line DC meets the line AB at F.

(b) Using a vector method, find $\overrightarrow{OF}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(7)

The point G lies on OB such that FG is parallel to AO.

(c) Using a vector method, find $\overrightarrow{OG}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(4)

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4. Question 10 · part a · 3 marks · Working and answer

Working and answer

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5. Question 10 · part a.i · 0 marks

2020 JAN · Paper 01

Further practice · Attempt part a.i only.

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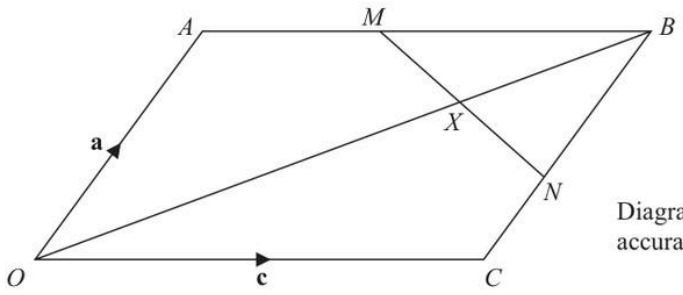


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the parallelogram $OABC$

$$\overrightarrow{OA} = \mathbf{a} \qquad \overrightarrow{OC} = \mathbf{c}$$

The midpoint of AB is M and the midpoint of BC is N.

The line OB intersects MN at the point X.

(a) Find in terms of **a** and **c**,

(i) $\overrightarrow{OB}$

(ii) $\overrightarrow{MN}$

(2)

Given $\overrightarrow{MX} = \lambda\overrightarrow{MN}$ and that $\overrightarrow{OX} = \mu\overrightarrow{OB}$,

(b) use a vector method to find the value of λ and the value of μ.

(8)

(c) Hence find, in its simplest form, the ratio

Area of quadrilateral $OXNC$: Area of parallelogram $OABC$.

(3)

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5. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 8 · part a · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a only.

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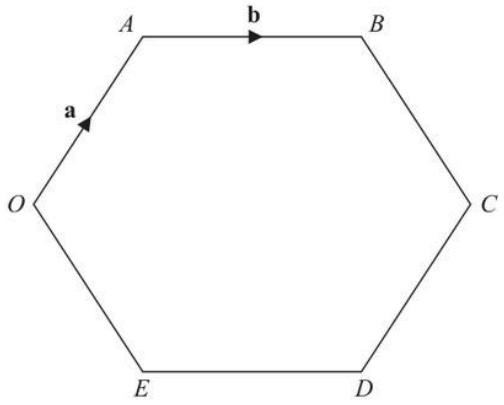


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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6. Question 8 · part a · 1 mark · Working and answer

Working and answer

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7. Question 8 · part b · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

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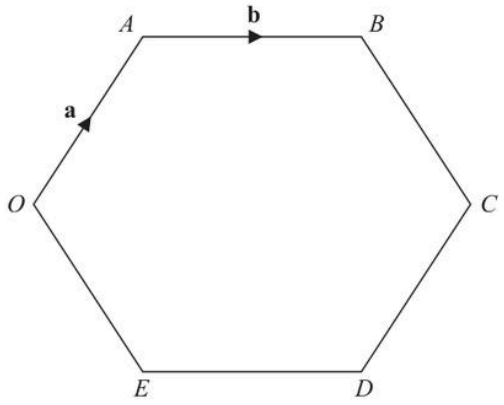


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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7. Question 8 · part b · 3 marks · Working and answer

Working and answer

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8. Question 11 · part a.i · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part a.i only.

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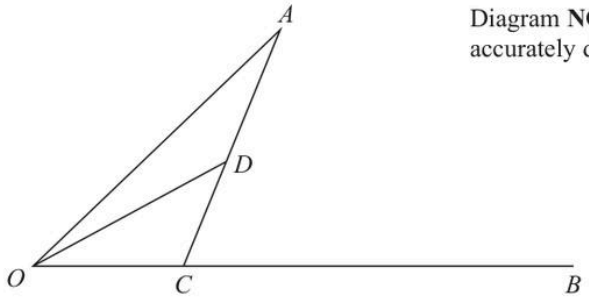


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C divides OB in the ratio $1 : 3$
The point D is the midpoint of AC

(a) Find, as a simplified expression in terms of **a** and **b**

(i) $\overrightarrow{AC}$ (ii) $\overrightarrow{OD}$ (iii) $\overrightarrow{BD}$

(5)

The point E is such that $\overrightarrow{OE} = \lambda\overrightarrow{OA}$

Given that E, D and B are collinear

(b) find the value of λ

(4)

Given that $\dfrac{\text{area } \Delta OAC}{\text{area } \Delta OEB} = \mu$

(c) find the value of μ

(4)

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8. Question 11 · part a.i · 5 marks · Working and answer

Working and answer

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9. Question 11 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

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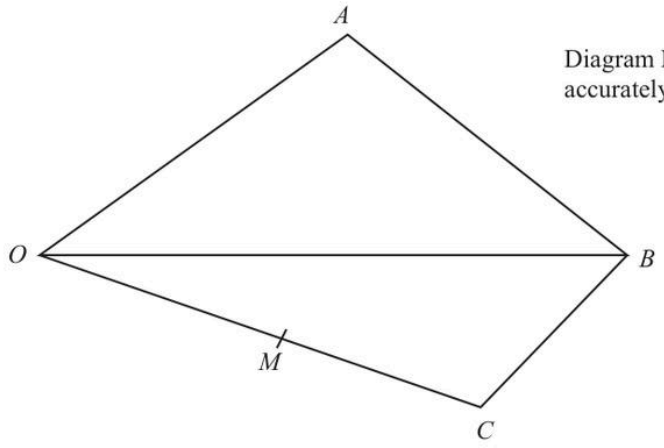


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows quadrilateral $OABC$ where

$$\overrightarrow{OA} = 4\mathbf{p} + 5\mathbf{q} \qquad \overrightarrow{OB} = 3\mathbf{p} + \mathbf{q} \qquad \overrightarrow{OC} = 2\mathbf{p} - 4\mathbf{q}$$

The point M is the midpoint of OC

(a) Find $\overrightarrow{MA}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$

(3)

The point N lies on OB such that M, N and A are collinear.

(b) Find the ratio $MN : NA$

(6)

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9. Question 11 · part a · 3 marks · Working and answer

Working and answer

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10. Question 9 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

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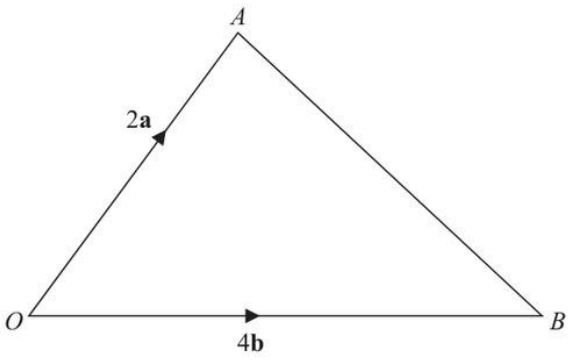


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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10. Question 9 · part b · 2 marks · Working and answer

Working and answer

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11. Question 9 · part c · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

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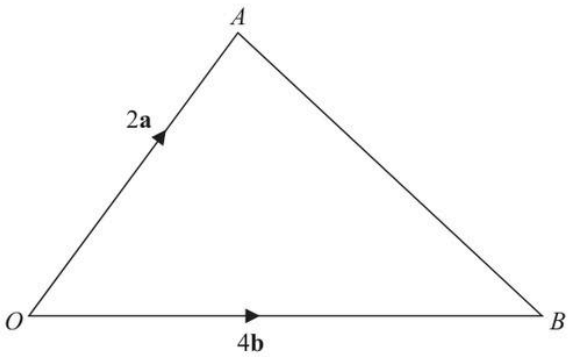


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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11. Question 9 · part c · 3 marks · Working and answer

Working and answer

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12. Question 10 · part a.ii · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

10

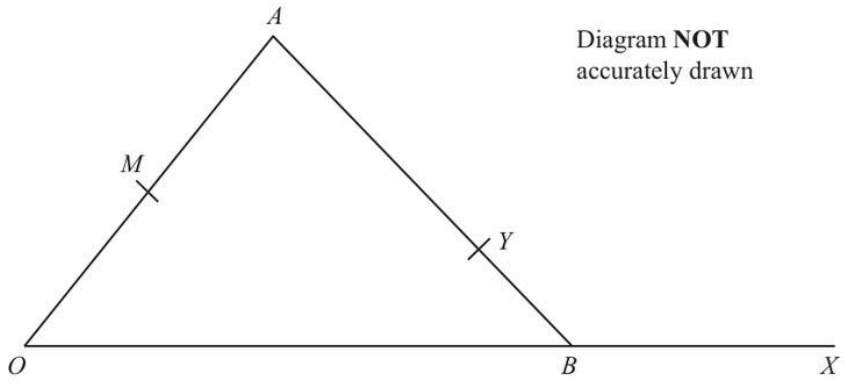


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = 2\mathbf{a}$, $\overrightarrow{OB} = 4\mathbf{b}$ and M is the midpoint of OA.

The point Y lies on AB such that $AY : YB = 3 : 1$

The point X lies on OB produced.

(a) Find as simplified expressions in terms of **a** and **b**

(i) $\overrightarrow{AB}$ (ii) $\overrightarrow{MY}$

(3)

The points M, Y and X are collinear.

(b) Find the ratio $OB : OX$

(5)

(c) Find the ratio of $(\text{Area } \Delta YBX) : (\text{Area } \Delta OAX)$

(3)

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12. Question 10 · part a.ii · 3 marks · Working and answer

Working and answer

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13. Question 10 · part a.ii · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part a.ii only.

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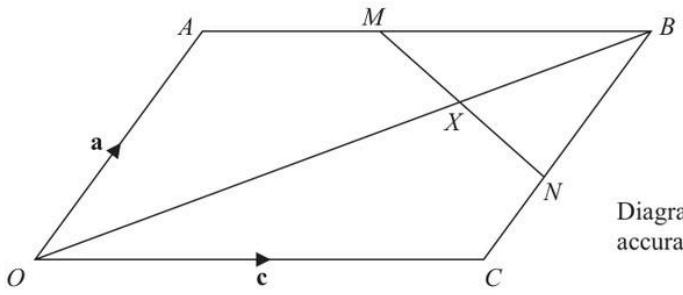


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the parallelogram $OABC$

$$\overrightarrow{OA} = \mathbf{a} \qquad \overrightarrow{OC} = \mathbf{c}$$

The midpoint of AB is M and the midpoint of BC is N.

The line OB intersects MN at the point X.

(a) Find in terms of **a** and **c**,

(i) $\overrightarrow{OB}$

(ii) $\overrightarrow{MN}$

(2)

Given $\overrightarrow{MX} = \lambda\overrightarrow{MN}$ and that $\overrightarrow{OX} = \mu\overrightarrow{OB}$,

(b) use a vector method to find the value of λ and the value of μ.

(8)

(c) Hence find, in its simplest form, the ratio

Area of quadrilateral $OXNC$: Area of parallelogram $OABC$.

(3)

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13. Question 10 · part a.ii · 2 marks · Working and answer

Working and answer

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14. Question 10 · part a.i · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

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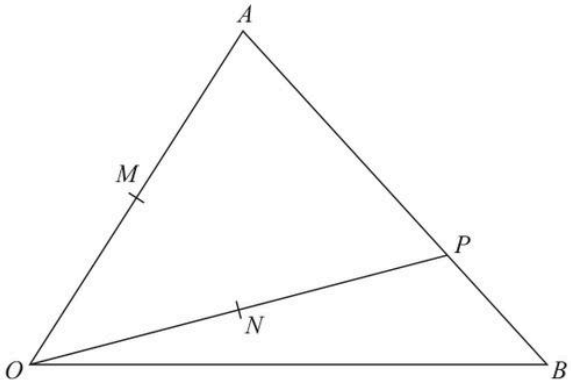


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows triangle OAB in which

$$\overrightarrow{OA} = \mathbf{a} \text{ and } \overrightarrow{OB} = \mathbf{b}$$

The point P lies on AB such that $AP : PB = 3 : 1$

The point M is the midpoint of OA and the point N is the midpoint of OP.

(a) Find, as simplified expressions in terms of **a** and **b**, the vector

(i) $\overrightarrow{OP}$ (ii) $\overrightarrow{MN}$

(4)

The point C lies on OB such that ANC is a straight line.

(b) Using a vector method, find the vector $\overrightarrow{OC}$ as a simplified expression in terms of **b**

(6)

Given that $\dfrac{\text{area of quadrilateral } AMNP}{\text{area of triangle } OAB} = K$

(c) find the exact value of K

(4)

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14. Question 10 · part a.i · 4 marks · Working and answer

Working and answer

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15. Question 8 · part c · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

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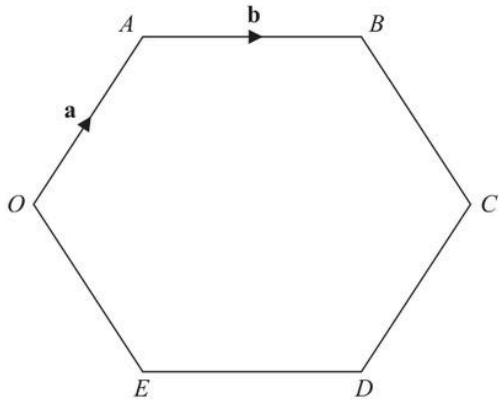


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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15. Question 8 · part c · 2 marks · Working and answer

Working and answer

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16. Question 11 · part a · 1 mark

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

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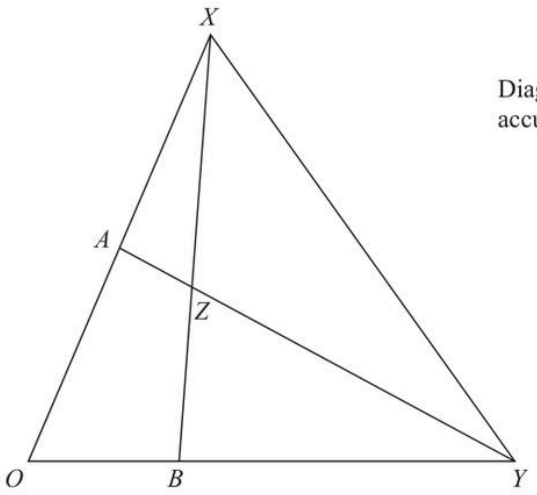


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangle OXY

$\overrightarrow{OX} = 2\mathbf{a}$ and $\overrightarrow{OY} = 3\mathbf{b}$

A is the midpoint of OX and B is the point on OY such that $OB : BY = 1 : 2$
The lines XB and AY intersect at Z.

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

(b) Using a vector method, find $\overrightarrow{OZ}$ as a simplified expression in terms of **a** and **b**

(9)

The point M on XY is such that O, Z and M are collinear.

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(3)

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P 6 2 2 8 3 A 0 3 4 4 0

16. Question 11 · part a · 1 mark · Working and answer

Working and answer

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17. Question 6 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 The points P, Q, R and S are the vertices of a quadrilateral $PQRS$ such that

$$\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} \qquad \overrightarrow{PR} = -\mathbf{i} + 18\mathbf{j} \qquad \overrightarrow{PS} = -3\mathbf{i} + 15\mathbf{j}$$

(a) Show that $PQRS$ is a parallelogram.

(4)

(b) Find a unit vector parallel to $\overrightarrow{QS}$ as a simplified expression in terms of **i** and **j**

(4)

The point T lies on QS such that $QT : TS = 5 : 8$

(c) Find $\overrightarrow{PT}$ as a simplified expression in terms of **i** and **j**

(2)

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17. Question 6 · part c · 2 marks

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Further practice · Attempt part c only.

Question 6 continued

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Turn over

17. Question 6 · part c · 2 marks

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Further practice · Attempt part c only.

Question 6 continued

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17. Question 6 · part c · 2 marks

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Further practice · Attempt part c only.



17. Question 6 · part c · 2 marks · Working and answer

Working and answer

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18. Question 6 · part c · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 Referred to a fixed origin O, the position vector of the point A is $(9\mathbf{a}+12\mathbf{b})$ and the position vector of the point B is $(2\mathbf{a}+p\mathbf{b})$ where $|\mathbf{a}|=|\mathbf{b}|=1$

Given that $|AB|=25$

(a) show that the two possible values of p are -12 and 36

(4)

For $p=-12$

(b) find a unit vector parallel to $\overrightarrow{OB}$
Give your answer in terms of **a** and **b** in its simplest form.

(2)

Point C lies on OB such that $OC:CB=2:3$

For $p=36$

(c) find $\overrightarrow{CA}$
Give your answer in terms of **a** and **b** in its simplest form.

(3)

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18. Question 6 · part c · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 9 marks)

18. Question 6 · part c · 3 marks · Working and answer

Working and answer

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Practice workbook · How long is a vector? Magnitudes and finding unknowns

Working out the length of a vector with Pythagoras on its components (or the cosine rule for non-perpendicular base vectors), and reversing this to find an unknown constant when a length is given.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-02

1. Question 4 · part b · 3 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part b only.

4 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$\overrightarrow{AB} = 7\mathbf{i} + p\mathbf{j}$ $\overrightarrow{AC} = 11\mathbf{i} - p\mathbf{j}$ $\overrightarrow{AD} = 4\mathbf{i} - 2p\mathbf{j}$

(a) Show that, for all values of p, $ABCD$ is a parallelogram.

(3)

Given that $|\overrightarrow{BD}| = 3\sqrt{10}$

(b) find the possible values of p.

(3)

Given that $p > 0$

(c) find a unit vector which is parallel to $\overrightarrow{BD}$.

(1)

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1. Question 4 · part b · 3 marks · Working and answer

Working and answer

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2. Question 1 · part b · 1 mark

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 The position vector of the point A is $(3\mathbf{i} - 2\mathbf{j})$, referred to a fixed origin O.

The point B is such that $\overrightarrow{AB} = (6\mathbf{i} + 8\mathbf{j})$

(a) Find the position vector of B as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find the magnitude of vector $\overrightarrow{AB}$

(1)

(c) Find a unit vector, in terms of $\mathbf{i}$ and $\mathbf{j}$, that is parallel to $\overrightarrow{AB}$

(2)

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2. Question 1 · part b · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = 5\mathbf{i} + 7\mathbf{j} \qquad \overrightarrow{AB} = a\mathbf{i} + 16\mathbf{j} \qquad \text{and} \qquad \left|\overrightarrow{OB}\right| = 5\sqrt{29}$$

(a) Find the possible values of a

(4)

Given that $a > 0$

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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3. Question 4 · part a · 4 marks · Working and answer

Working and answer

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4. Question 3 · part a · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

3 O, A and B are fixed points such that

$$\overrightarrow{OA} = (p\mathbf{i} - 4\mathbf{j}) \qquad \overrightarrow{OB} = \mathbf{i} + (2p + 1)\mathbf{j}$$

Given that $\sqrt{2}\left|\overrightarrow{OA}\right| = \left|\overrightarrow{OB}\right|$ and $p > 0$

(a) find the value of p

(4)

Using this value of p

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(5)

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4. Question 3 · part a · 4 marks · Working and answer

Working and answer

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5. Question 3 · part a · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

3 O, A and B are fixed points such that

$$\left|\overrightarrow{OA}\right| = 3\sqrt{5} \qquad \overrightarrow{AB} = \mathbf{i} + 3a\mathbf{j} \qquad \overrightarrow{OB} = 7\mathbf{i} + 2a\mathbf{j}$$

Given that $a > 0$

(a) find the value of a

(4)

(b) Hence find a unit vector parallel to $\overrightarrow{OA}$

(2)

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5. Question 3 · part a · 4 marks · Working and answer

Working and answer

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6. Question 6 · part a · 4 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 Referred to a fixed origin O, the position vector of the point A is $(9\mathbf{a}+12\mathbf{b})$ and the position vector of the point B is $(2\mathbf{a}+p\mathbf{b})$ where $|\mathbf{a}|=|\mathbf{b}|=1$

Given that $|AB|=25$

(a) show that the two possible values of p are -12 and 36

(4)

For $p=-12$

(b) find a unit vector parallel to $\overrightarrow{OB}$
Give your answer in terms of **a** and **b** in its simplest form.

(2)

Point C lies on OB such that $OC:CB=2:3$

For $p=36$

(c) find $\overrightarrow{CA}$
Give your answer in terms of **a** and **b** in its simplest form.

(3)

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6. Question 6 · part a · 4 marks · Working and answer

Working and answer

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7. Question 8 · part a.ii · 3 marks

2024 NOV · Paper 02

Further practice · Attempt part a.ii only.

8

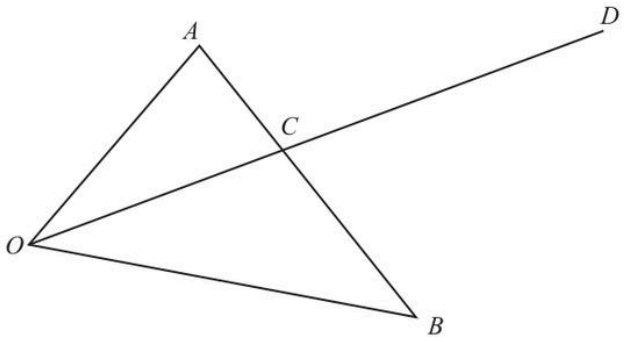


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$ **(3)**

(b) Find the area of triangle AOB **(4)**

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of **a** and **b** **(5)**

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P 7 3 5 8 7 A 0 2 0 3 2

7. Question 8 · part a.ii · 3 marks · Working and answer

Working and answer

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Practice workbook · Unit vectors: same direction, length one

Finding a vector of length one in the direction of a given vector by dividing the vector by its magnitude, including working backwards from a given unit vector to find unknown constants.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-03

1. Question 6 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Worked-example retry · Attempt part b only.

6 The points P, Q, R and S are the vertices of a quadrilateral $PQRS$ such that

$$\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} \qquad \overrightarrow{PR} = -\mathbf{i} + 18\mathbf{j} \qquad \overrightarrow{PS} = -3\mathbf{i} + 15\mathbf{j}$$

(a) Show that $PQRS$ is a parallelogram.

(4)

(b) Find a unit vector parallel to $\overrightarrow{QS}$ as a simplified expression in terms of **i** and **j**

(4)

The point T lies on QS such that $QT : TS = 5 : 8$

(c) Find $\overrightarrow{PT}$ as a simplified expression in terms of **i** and **j**

(2)

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1. Question 6 · part b · 4 marks · Working and answer

Working and answer

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2. Question 7 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 O, A, B and C are fixed points such that

$$\overrightarrow{OA} = 8\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OB} = 15\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OC} = 8\mathbf{i} + \mathbf{j}$$

(a) Find $\overrightarrow{BC}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector parallel to $\overrightarrow{BC}$

(2)

The point M is the midpoint of OA and the point N lies on OB such that $ON : NB = 1 : 2$

(c) Show that the points M, N and C are collinear.

(4)

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2. Question 7 · part b · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

1 Referred to a fixed origin O, the point A has position vector $(4\mathbf{i} + 3\mathbf{j})$ and the point B has position vector $(\mathbf{i} + 7\mathbf{j})$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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(Total for Question 1 is 4 marks)

3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 1 · part c · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

1 The position vector of the point A is $(3\mathbf{i} - 2\mathbf{j})$, referred to a fixed origin O.

The point B is such that $\overrightarrow{AB} = (6\mathbf{i} + 8\mathbf{j})$

(a) Find the position vector of B as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find the magnitude of vector $\overrightarrow{AB}$

(1)

(c) Find a unit vector, in terms of $\mathbf{i}$ and $\mathbf{j}$, that is parallel to $\overrightarrow{AB}$

(2)

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4. Question 1 · part c · 2 marks

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Further practice · Attempt part c only.

Question 1 continued

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(Total for Question 1 is 5 marks)

4. Question 1 · part c · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = 5\mathbf{i} + 7\mathbf{j} \qquad \overrightarrow{AB} = a\mathbf{i} + 16\mathbf{j} \qquad \text{and} \qquad \left|\overrightarrow{OB}\right| = 5\sqrt{29}$$

(a) Find the possible values of a

(4)

Given that $a > 0$

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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5. Question 4 · part b · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 6 marks)

5. Question 4 · part b · 2 marks · Working and answer

Working and answer

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

4 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$\overrightarrow{AB} = 7\mathbf{i} + p\mathbf{j}$ $\overrightarrow{AC} = 11\mathbf{i} - p\mathbf{j}$ $\overrightarrow{AD} = 4\mathbf{i} - 2p\mathbf{j}$

(a) Show that, for all values of p, $ABCD$ is a parallelogram.

(3)

Given that $|\overrightarrow{BD}| = 3\sqrt{10}$

(b) find the possible values of p.

(3)

Given that $p > 0$

(c) find a unit vector which is parallel to $\overrightarrow{BD}$.

(1)

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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6. Question 4 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 7 marks)

6. Question 4 · part c · 1 mark · Working and answer

Working and answer

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7. Question 3 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

3 O, A and B are fixed points such that

$$\overrightarrow{OA} = (p\mathbf{i} - 4\mathbf{j}) \qquad \overrightarrow{OB} = \mathbf{i} + (2p + 1)\mathbf{j}$$

Given that $\sqrt{2}\left|\overrightarrow{OA}\right| = \left|\overrightarrow{OB}\right|$ and $p > 0$

(a) find the value of p

(4)

Using this value of p

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(5)

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7. Question 3 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 9 marks)

7. Question 3 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part b · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

3 O, A and B are fixed points such that

$$\left|\overrightarrow{OA}\right| = 3\sqrt{5} \qquad \overrightarrow{AB} = \mathbf{i} + 3a\mathbf{j} \qquad \overrightarrow{OB} = 7\mathbf{i} + 2a\mathbf{j}$$

Given that $a > 0$

(a) find the value of a

(4)

(b) Hence find a unit vector parallel to $\overrightarrow{OA}$

(2)

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8. Question 3 · part b · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

8. Question 3 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 10 · part whole · 10 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

10 O, A and B are fixed points such that

$\overrightarrow{OA} = (b + 1)\mathbf{i} + b\mathbf{j}$

$\overrightarrow{AB} = 3\mathbf{i}$

The unit vector parallel to $\overrightarrow{OB}$ is $\frac{\sqrt{17}}{34}[(3a + 2)\mathbf{i} + b\mathbf{j}]$

Given that a and b are constants where $a > 0$ and $b > 0$

find the exact value of

(i) a

(ii) b

(10)

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9. Question 10 · part whole · 10 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

Question 10 continued

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9. Question 10 · part whole · 10 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

Question 10 continued

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9. Question 10 · part whole · 10 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

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Question 10 continued

OLevels.com SCHOOL

(Total for Question 10 is 10 marks)

9. Question 10 · part whole · 10 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 4 · part c · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = p\mathbf{i} + 2p\mathbf{j} \qquad \overrightarrow{OB} = 5\mathbf{i} + 9p\mathbf{j}$$

Given that $\overrightarrow{AB}$ is parallel to $(\mathbf{i} - 2\mathbf{j})$

(a) find the value of p

(6)

(b) Hence find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(c) Find a unit vector parallel to $\overrightarrow{OA}$

Give your answer in the form $\frac{\sqrt{a}}{5}(b\mathbf{i} + c\mathbf{j})$ where a, b and c are integers to be found.

(4)

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P 7 1 8 1 8 A 0 8 2 8

10. Question 4 · part c · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 12 marks)

10. Question 4 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 6 · part b · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 Referred to a fixed origin O, the position vector of the point A is $(9\mathbf{a}+12\mathbf{b})$ and the position vector of the point B is $(2\mathbf{a}+p\mathbf{b})$ where $|\mathbf{a}|=|\mathbf{b}|=1$

Given that $|AB|=25$

(a) show that the two possible values of p are -12 and 36

(4)

For $p=-12$

(b) find a unit vector parallel to $\overrightarrow{OB}$
Give your answer in terms of **a** and **b** in its simplest form.

(2)

Point C lies on OB such that $OC:CB=2:3$

For $p=36$

(c) find $\overrightarrow{CA}$
Give your answer in terms of **a** and **b** in its simplest form.

(3)

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P 7 6 5 0 7 A 0 1 2 3 2

11. Question 6 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 9 · part b · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

9 The points A, B, C and D are the vertices of a quadrilateral where

$$\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b} \qquad \overrightarrow{AC} = 7\mathbf{a} + 7\mathbf{b} \qquad \overrightarrow{AD} = 10\mathbf{a} + 6\mathbf{b}$$

(a) Show that $ABCD$ is a trapezium.

(4)

Given that $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$ and that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other

(b) find a unit vector parallel to $\overrightarrow{BD}$

(4)

The point Y lies on CD such that $CY : YD = 1 : 2$

The lines AC and BY intersect at the point X

(c) (i) Use a vector method to find the ratio $BX : XY$

(4)

(ii) Hence find, in its simplest form, the ratio

area of triangle CXY : area of trapezium $ABCD$

(4)

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12. Question 9 · part b · 4 marks · Working and answer

Working and answer

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Practice workbook · Proving shapes and straight lines with vectors

Using the fact that parallel vectors are scalar multiples of each other to prove lines parallel, points collinear or a quadrilateral a parallelogram or trapezium, or to find an unknown that makes two vectors parallel.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-06

1. Question 4 · part a · 3 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part a only.

4 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$\overrightarrow{AB} = 7\mathbf{i} + p\mathbf{j}$ $\overrightarrow{AC} = 11\mathbf{i} - p\mathbf{j}$ $\overrightarrow{AD} = 4\mathbf{i} - 2p\mathbf{j}$

(a) Show that, for all values of p, $ABCD$ is a parallelogram.

(3)

Given that $|\overrightarrow{BD}| = 3\sqrt{10}$

(b) find the possible values of p.

(3)

Given that $p > 0$

(c) find a unit vector which is parallel to $\overrightarrow{BD}$.

(1)

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P 5 9 9 3 9 A 0 1 0 3 6

1. Question 4 · part a · 3 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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1 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$$\overrightarrow{AB} = 3\mathbf{a} + 2\mathbf{b} \qquad \overrightarrow{AD} = 2\mathbf{a} - 5\mathbf{b} \qquad \overrightarrow{AC} = 5\mathbf{a} - 3\mathbf{b}$$

Show that $ABCD$ is a parallelogram.

(4)

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(Total for Question 1 is 4 marks)

2. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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3. Question 10 · part a · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

10 The points A, B, C and D are the vertices of a quadrilateral such that

$$\overrightarrow{AB} = 3\mathbf{a} + 4\mathbf{b} \quad \overrightarrow{AC} = 7\mathbf{a} + 9\mathbf{b} \quad \overrightarrow{AD} = 4\mathbf{a} + 5\mathbf{b}$$

(a) Show that $ABCD$ is a parallelogram.

(3)

BC is extended to the point E such that BCE is a straight line.

Point F lies on CD such that $CF : FD = 1 : 2$

Given that A, F and E are collinear,

(b) find the vector $\overrightarrow{AE}$ in the form $X\mathbf{a} + Y\mathbf{b}$ where X and Y are rational numbers to be found.

(8)

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3. Question 10 · part a · 3 marks · Working and answer

Working and answer

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4. Question 9 · part a · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

9 The points A, B, C and D are the vertices of a quadrilateral where

$$\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b} \qquad \overrightarrow{AC} = 7\mathbf{a} + 7\mathbf{b} \qquad \overrightarrow{AD} = 10\mathbf{a} + 6\mathbf{b}$$

(a) Show that $ABCD$ is a trapezium.

(4)

Given that $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$ and that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other

(b) find a unit vector parallel to $\overrightarrow{BD}$

(4)

The point Y lies on CD such that $CY : YD = 1 : 2$

The lines AC and BY intersect at the point X

(c) (i) Use a vector method to find the ratio $BX : XY$

(4)

(ii) Hence find, in its simplest form, the ratio

area of triangle CXY : area of trapezium $ABCD$

(4)

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20

P 7 8 9 6 5 A 0 2 0 3 2

4. Question 9 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 10 · part b · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

10

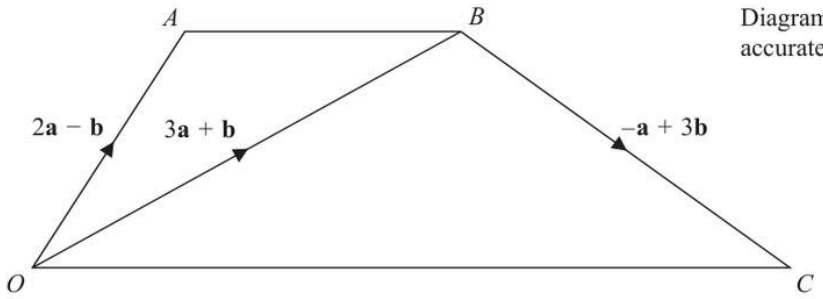


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows quadrilateral $OABC$ with

$$\overrightarrow{OA} = 2\mathbf{a} - \mathbf{b} \qquad \overrightarrow{OB} = 3\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = -\mathbf{a} + 3\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**.

(2)

(b) Prove that $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$

(2)

The diagonals, OB and AC, intersect at the point X.

(c) Using a vector method find the ratio $AX : XC$

(7)

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5. Question 10 · part b · 2 marks · Working and answer

Working and answer

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6. Question 9 · part a · 4 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9

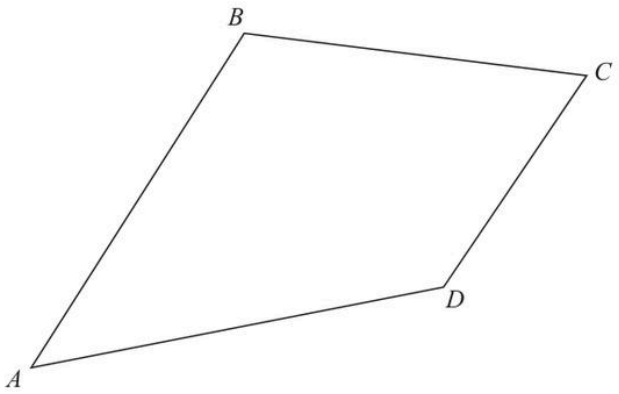


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows quadrilateral $ABCD$ such that

$$\overrightarrow{AD} = 2\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = \frac{1}{3}\mathbf{b} \qquad \overrightarrow{BD} = -4\mathbf{a} - \mathbf{b}$$

(a) Prove that $\overrightarrow{AB}$ is parallel to $\overrightarrow{DC}$ **(4)**

The diagonals, AC and BD, of the quadrilateral intersect at the point Y.

(b) Using a vector method, find $\overrightarrow{AY}$ as a simplified expression in terms of **a** and **b** **(6)**

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P 7 1 6 4 1 A 0 2 4 3 6

6. Question 9 · part a · 4 marks · Working and answer

Working and answer

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7. Question 6 · part a · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 The points P, Q, R and S are the vertices of a quadrilateral $PQRS$ such that

$$\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} \qquad \overrightarrow{PR} = -\mathbf{i} + 18\mathbf{j} \qquad \overrightarrow{PS} = -3\mathbf{i} + 15\mathbf{j}$$

(a) Show that $PQRS$ is a parallelogram.

(4)

(b) Find a unit vector parallel to $\overrightarrow{QS}$ as a simplified expression in terms of **i** and **j**

(4)

The point T lies on QS such that $QT : TS = 5 : 8$

(c) Find $\overrightarrow{PT}$ as a simplified expression in terms of **i** and **j**

(2)

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7. Question 6 · part a · 4 marks · Working and answer

Working and answer

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8. Question 7 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 O, A, B and C are fixed points such that

$$\overrightarrow{OA} = 8\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OB} = 15\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OC} = 8\mathbf{i} + \mathbf{j}$$

(a) Find $\overrightarrow{BC}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector parallel to $\overrightarrow{BC}$

(2)

The point M is the midpoint of OA and the point N lies on OB such that $ON : NB = 1 : 2$

(c) Show that the points M, N and C are collinear.

(4)

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8. Question 7 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

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Question 7 continued

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8. Question 7 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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8. Question 7 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.



Question 7 continued

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(Total for Question 7 is 8 marks)

8. Question 7 · part c · 4 marks · Working and answer

Working and answer

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9. Question 11 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

11

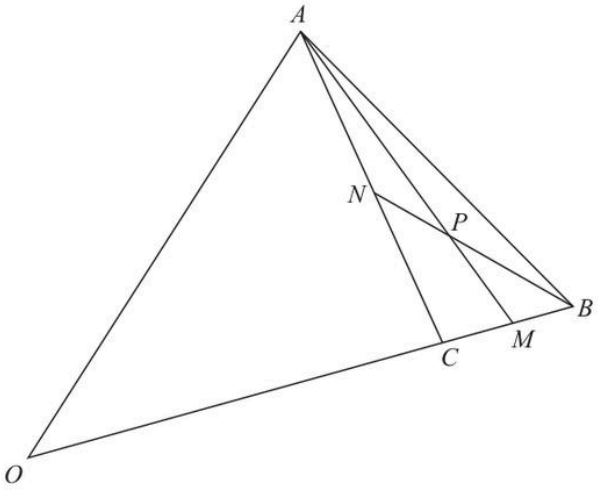


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C lies on OB such that $OC : CB = 2 : 1$
The point M is the midpoint of CB and the point N is the midpoint of AC
The lines AM and NB intersect at the point P

(a) Using a vector method, find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(9)

The point Q is the midpoint of AB

(b) Using a vector method, show that C, P and Q are collinear.

(4)

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Olevels.com SCHOOL

9. Question 11 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 11 continued

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9. Question 11 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 13 marks)

9. Question 11 · part b · 4 marks · Working and answer

Working and answer

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10. Question 4 · part a · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = p\mathbf{i} + 2p\mathbf{j} \qquad \overrightarrow{OB} = 5\mathbf{i} + 9p\mathbf{j}$$

Given that $\overrightarrow{AB}$ is parallel to $(\mathbf{i} - 2\mathbf{j})$

(a) find the value of p

(6)

(b) Hence find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(c) Find a unit vector parallel to $\overrightarrow{OA}$

Give your answer in the form $\frac{\sqrt{a}}{5}(b\mathbf{i} + c\mathbf{j})$ where a, b and c are integers to be found.

(4)

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OLevels.com SCHOOL

P 7 1 8 1 8 A 0 8 2 8

10. Question 4 · part a · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Where lines cross: two routes and comparing coefficients

Finding the position of a point where two lines cross, or a point fixed by a collinear or parallel condition, by writing its vector in two ways with unknown scalars and comparing coefficients of the two non-parallel base vectors.

20 question parts: 1 worked-example retry and 19 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-07

1. Question 11 · part b · 7 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part b only.

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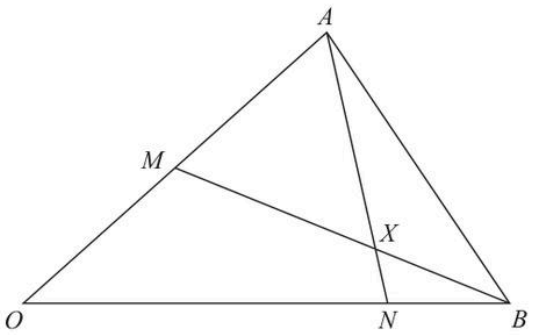


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

M is the midpoint of OA.

N is the point on OB such that $ON : NB = 3 : 1$

The lines AN and BM intersect at the point X.

(a) Find expressions, in terms of **a** and **b**, for

(i) $\overrightarrow{AN}$ (ii) $\overrightarrow{BM}$

(3)

(b) Using a vector method, find $AX : XN$

(7)

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P 7 1 8 1 7 A 0 3 0 3 2

1. Question 11 · part b · 7 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 11 continued

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1. Question 11 · part b · 7 marks

2023 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 10 marks)

1. Question 11 · part b · 7 marks · Working and answer

Working and answer

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2. Question 11 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

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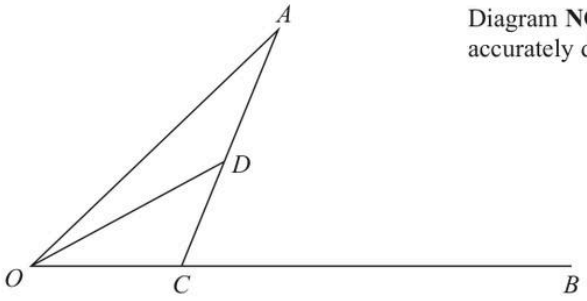


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C divides OB in the ratio $1 : 3$
The point D is the midpoint of AC

(a) Find, as a simplified expression in terms of **a** and **b**

(i) $\overrightarrow{AC}$ (ii) $\overrightarrow{OD}$ (iii) $\overrightarrow{BD}$

(5)

The point E is such that $\overrightarrow{OE} = \lambda\overrightarrow{OA}$

Given that E, D and B are collinear

(b) find the value of λ

(4)

Given that $\dfrac{\text{area } \Delta OAC}{\text{area } \Delta OEB} = \mu$

(c) find the value of μ

(4)

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30

P 6 2 2 8 0 A 0 3 0 3 2

2. Question 11 · part b · 4 marks · Working and answer

Working and answer

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3. Question 11 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

11

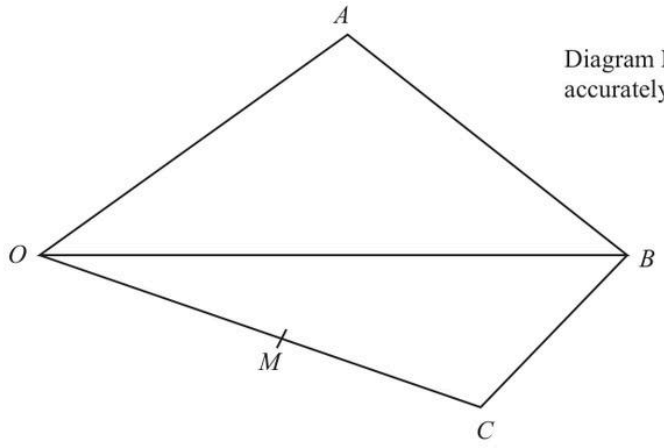


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows quadrilateral $OABC$ where

$$\overrightarrow{OA} = 4\mathbf{p} + 5\mathbf{q} \quad \overrightarrow{OB} = 3\mathbf{p} + \mathbf{q} \quad \overrightarrow{OC} = 2\mathbf{p} - 4\mathbf{q}$$

The point M is the midpoint of OC

(a) Find $\overrightarrow{MA}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$

(3)

The point N lies on OB such that M, N and A are collinear.

(b) Find the ratio $MN : NA$

(6)

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P 7 3 5 8 6 A 0 3 0 3 2

3. Question 11 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 11 continued

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3. Question 11 · part b · 6 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

Question 11 continued

OLevels.com SCHOOL

(Total for Question 11 is 9 marks)

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3. Question 11 · part b · 6 marks · Working and answer

Working and answer

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4. Question 11 · part a · 9 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

11

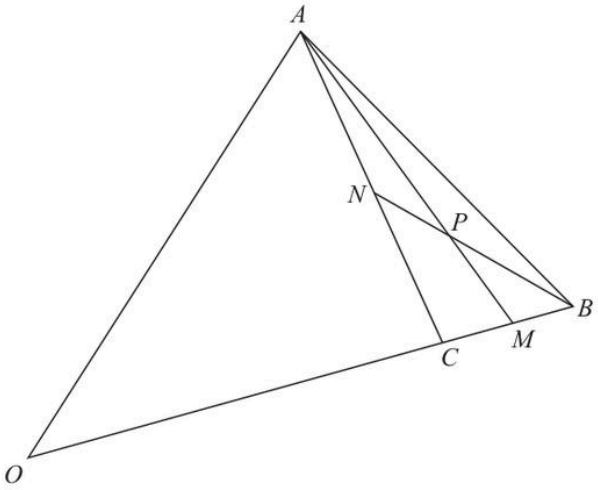


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C lies on OB such that $OC : CB = 2 : 1$
The point M is the midpoint of CB and the point N is the midpoint of AC
The lines AM and NB intersect at the point P

(a) Using a vector method, find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(9)

The point Q is the midpoint of AB

(b) Using a vector method, show that C, P and Q are collinear.

(4)

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Olevels.com SCHOOL

4. Question 11 · part a · 9 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

8

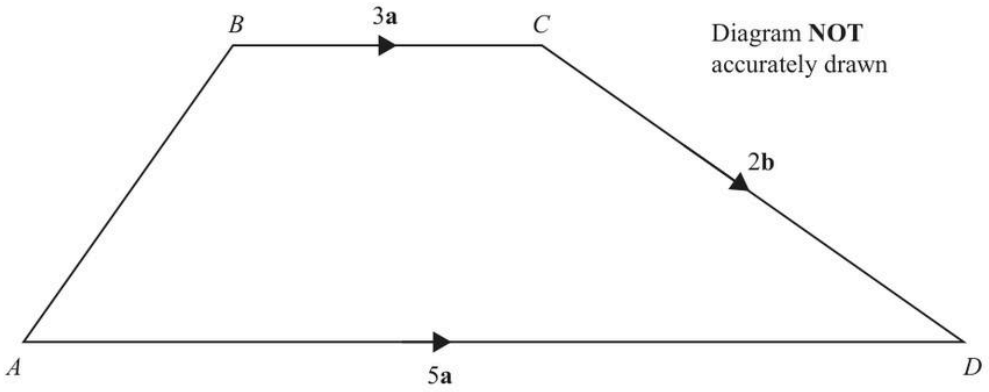


Figure 3

Figure 3 shows a trapezium $ABCD$

$\overrightarrow{BC} = 3\mathbf{a}$ $\overrightarrow{AD} = 5\mathbf{a}$ $\overrightarrow{CD} = 2\mathbf{b}$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

The diagonals BD and AC intersect at point X where $\overrightarrow{BX} = k\overrightarrow{BD}$

(b) Using a vector method, find the value of k

(5)

(c) Find the ratio of the area of triangle CXD : area of the trapezium $ABCD$

(4)

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P 7 4 0 9 9 A 0 1 8 3 2

5. Question 8 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 9 · part d · 6 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

9

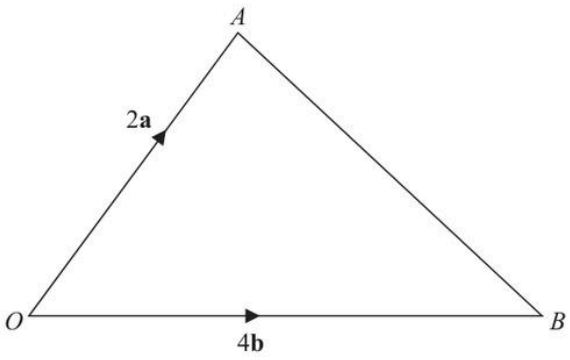


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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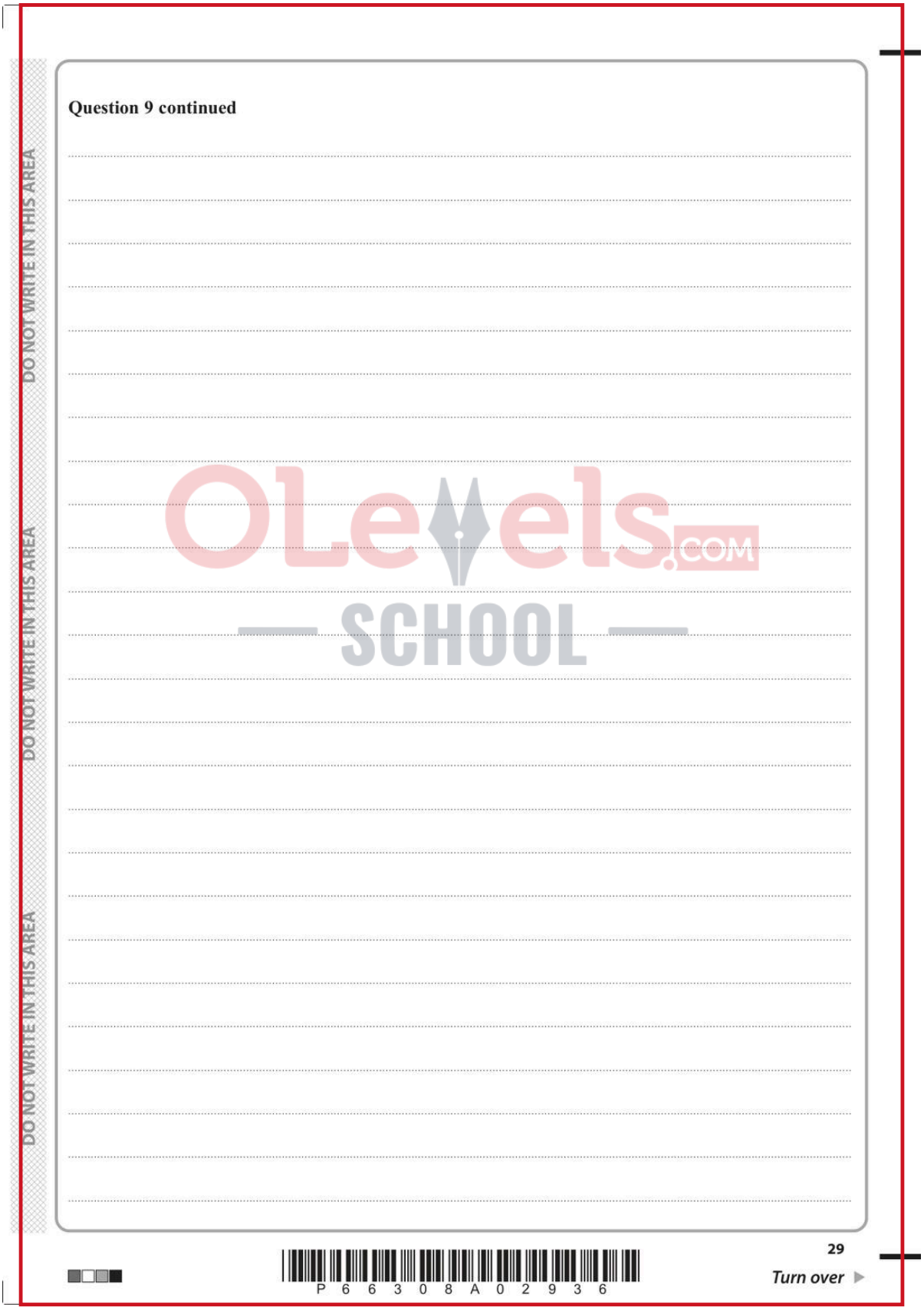
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OLevels.COM — SCHOOL —

6. Question 9 · part d · 6 marks

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Further practice · Attempt part d only.



Question 9 continued

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P 6 6 3 0 8 A 0 2 9 3 6

29

Turn over

6. Question 9 · part d · 6 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.



Question 9 continued

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P 6 6 3 0 8 A 0 3 0 3 6

6. Question 9 · part d · 6 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 13 marks)

6. Question 9 · part d · 6 marks · Working and answer

Working and answer

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7. Question 8 · part b · 4 marks

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Further practice · Attempt part b only.

8

Diagram **NOT** accurately drawn

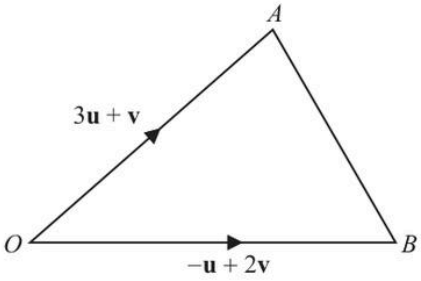


Figure 2

Figure 2 shows triangle OAB with

$$\overrightarrow{OA} = 3\mathbf{u} + \mathbf{v} \qquad \overrightarrow{OB} = -\mathbf{u} + 2\mathbf{v}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (2)

Point C is such that AC is parallel to OB

Given that $\overrightarrow{OC} = \mu\mathbf{v}$

(b) find the value of μ (4)

The lines OC and AB intersect at point X

Point D lies on $\overrightarrow{OC}$ such that

area of triangle BOX : area of triangle $BXD = 2 : 3$

Using a vector method,

(c) find $\overrightarrow{BD}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (5)

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P 7 2 8 6 7 A 0 1 8 3 2

7. Question 8 · part b · 4 marks · Working and answer

Working and answer

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8. Question 10 · part c · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

10

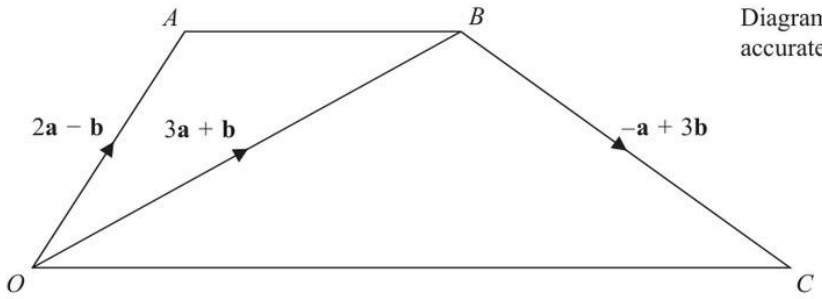


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows quadrilateral $OABC$ with

$$\overrightarrow{OA} = 2\mathbf{a} - \mathbf{b} \qquad \overrightarrow{OB} = 3\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = -\mathbf{a} + 3\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**.

(2)

(b) Prove that $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$

(2)

The diagonals, OB and AC, intersect at the point X.

(c) Using a vector method find the ratio $AX : XC$

(7)

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P 6 0 4 7 5 A 0 2 8 3 6

8. Question 10 · part c · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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8. Question 10 · part c · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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8. Question 10 · part c · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 11 marks)

8. Question 10 · part c · 7 marks · Working and answer

Working and answer

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9. Question 10 · part b · 7 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

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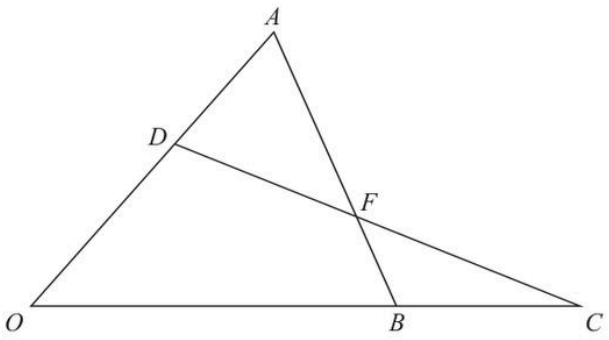


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle OAB and triangle OCD.

$$\overrightarrow{OA} = 5\mathbf{p} \quad \overrightarrow{AB} = 3\mathbf{q} \quad \overrightarrow{OC} = \frac{3}{2}\overrightarrow{OB} \quad \overrightarrow{OD} = \frac{3}{5}\overrightarrow{OA}$$

(a) Find $\overrightarrow{DC}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(3)

The line DC meets the line AB at F.

(b) Using a vector method, find $\overrightarrow{OF}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(7)

The point G lies on OB such that FG is parallel to AO.

(c) Using a vector method, find $\overrightarrow{OG}$ as a simplified expression in terms of $\mathbf{p}$ and $\mathbf{q}$.

(4)

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9. Question 10 · part b · 7 marks · Working and answer

Working and answer

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10. Question 10 · part c · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

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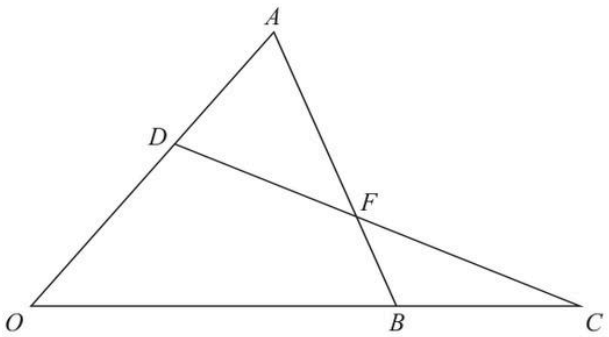


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle OAB and triangle OCD.

$$\overrightarrow{OA} = 5\mathbf{p} \quad \overrightarrow{AB} = 3\mathbf{q} \quad \overrightarrow{OC} = \frac{3}{2}\overrightarrow{OB} \quad \overrightarrow{OD} = \frac{3}{5}\overrightarrow{OA}$$

(a) Find $\overrightarrow{DC}$ as a simplified expression in terms of **p** and **q**.

(3)

The line DC meets the line AB at F.

(b) Using a vector method, find $\overrightarrow{OF}$ as a simplified expression in terms of **p** and **q**.

(7)

The point G lies on OB such that FG is parallel to AO.

(c) Using a vector method, find $\overrightarrow{OG}$ as a simplified expression in terms of **p** and **q**.

(4)

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10. Question 10 · part c · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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10. Question 10 · part c · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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10. Question 10 · part c · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 14 marks)

10. Question 10 · part c · 4 marks · Working and answer

Working and answer

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11. Question 9 · part b · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

9

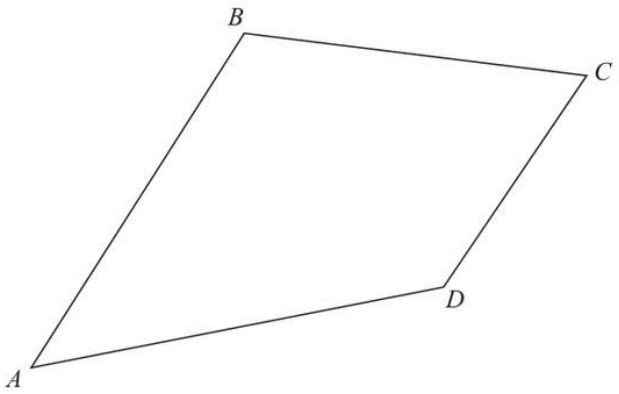


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows quadrilateral $ABCD$ such that

$$\overrightarrow{AD} = 2\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = \frac{1}{3}\mathbf{b} \qquad \overrightarrow{BD} = -4\mathbf{a} - \mathbf{b}$$

(a) Prove that $\overrightarrow{AB}$ is parallel to $\overrightarrow{DC}$ **(4)**

The diagonals, AC and BD, of the quadrilateral intersect at the point Y.

(b) Using a vector method, find $\overrightarrow{AY}$ as a simplified expression in terms of **a** and **b** **(6)**

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11. Question 9 · part b · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 9 continued

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11. Question 9 · part b · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 9 continued

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11. Question 9 · part b · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 9 continued

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(Total for Question 9 is 10 marks)

11. Question 9 · part b · 6 marks · Working and answer

Working and answer

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12. Question 10 · part b · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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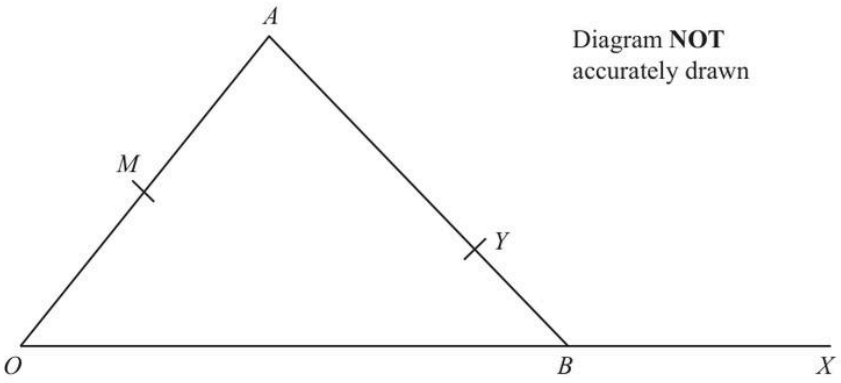


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = 2\mathbf{a}$, $\overrightarrow{OB} = 4\mathbf{b}$ and M is the midpoint of OA.

The point Y lies on AB such that $AY : YB = 3 : 1$

The point X lies on OB produced.

(a) Find as simplified expressions in terms of **a** and **b**

(i) $\overrightarrow{AB}$ (ii) $\overrightarrow{MY}$

(3)

The points M, Y and X are collinear.

(b) Find the ratio $OB : OX$

(5)

(c) Find the ratio of $(\text{Area } \Delta YBX) : (\text{Area } \Delta OAX)$

(3)

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P 7 4 0 9 8 A 0 3 0 3 2

12. Question 10 · part b · 5 marks · Working and answer

Working and answer

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13. Question 10 · part b · 8 marks

2020 JAN · Paper 01

Further practice · Attempt part b only.

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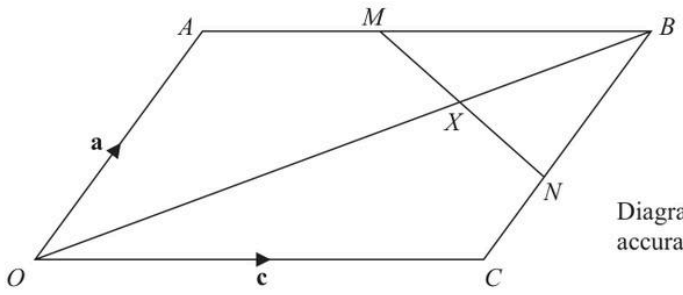


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the parallelogram $OABC$

$$\overrightarrow{OA} = \mathbf{a} \qquad \overrightarrow{OC} = \mathbf{c}$$

The midpoint of AB is M and the midpoint of BC is N.

The line OB intersects MN at the point X.

(a) Find in terms of **a** and **c**,

(i) $\overrightarrow{OB}$

(ii) $\overrightarrow{MN}$

(2)

Given $\overrightarrow{MX} = \lambda\overrightarrow{MN}$ and that $\overrightarrow{OX} = \mu\overrightarrow{OB}$,

(b) use a vector method to find the value of λ and the value of μ.

(8)

(c) Hence find, in its simplest form, the ratio

Area of quadrilateral $OXNC$: Area of parallelogram $OABC$.

(3)

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P 5 9 9 3 8 A 0 3 0 3 6

13. Question 10 · part b · 8 marks · Working and answer

Working and answer

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14. Question 10 · part b · 6 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

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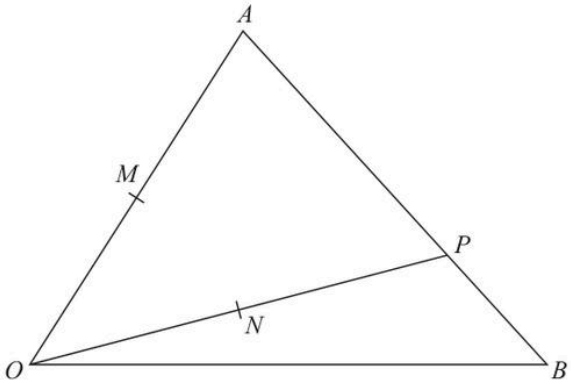


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows triangle OAB in which

$$\overrightarrow{OA} = \mathbf{a} \text{ and } \overrightarrow{OB} = \mathbf{b}$$

The point P lies on AB such that $AP : PB = 3 : 1$

The point M is the midpoint of OA and the point N is the midpoint of OP.

(a) Find, as simplified expressions in terms of **a** and **b**, the vector

(i) $\overrightarrow{OP}$ (ii) $\overrightarrow{MN}$

(4)

The point C lies on OB such that ANC is a straight line.

(b) Using a vector method, find the vector $\overrightarrow{OC}$ as a simplified expression in terms of **b**

(6)

Given that $\dfrac{\text{area of quadrilateral } AMNP}{\text{area of triangle } OAB} = K$

(c) find the exact value of K

(4)

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P 7 1 6 6 1 A 0 3 0 3 2

14. Question 10 · part b · 6 marks · Working and answer

Working and answer

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15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

10 The points A, B, C and D are the vertices of a quadrilateral such that

$$\overrightarrow{AB} = 3\mathbf{a} + 4\mathbf{b} \quad \overrightarrow{AC} = 7\mathbf{a} + 9\mathbf{b} \quad \overrightarrow{AD} = 4\mathbf{a} + 5\mathbf{b}$$

(a) Show that $ABCD$ is a parallelogram.

(3)

BC is extended to the point E such that BCE is a straight line.

Point F lies on CD such that $CF : FD = 1 : 2$

Given that A, F and E are collinear,

(b) find the vector $\overrightarrow{AE}$ in the form $X\mathbf{a} + Y\mathbf{b}$ where X and Y are rational numbers to be found.

(8)

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P 7 6 5 0 8 A 0 2 6 3 2

15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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Question 10 continued

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P 7 6 5 0 8 A 0 2 7 3 2

15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

Question 10 continued

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15. Question 10 · part b · 8 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 11 marks)

15. Question 10 · part b · 8 marks · Working and answer

Working and answer

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16. Question 9 · part c.i · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c.i only.

9 The points A, B, C and D are the vertices of a quadrilateral where

$$\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b} \qquad \overrightarrow{AC} = 7\mathbf{a} + 7\mathbf{b} \qquad \overrightarrow{AD} = 10\mathbf{a} + 6\mathbf{b}$$

(a) Show that $ABCD$ is a trapezium.

(4)

Given that $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$ and that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other

(b) find a unit vector parallel to $\overrightarrow{BD}$

(4)

The point Y lies on CD such that $CY : YD = 1 : 2$

The lines AC and BY intersect at the point X

(c) (i) Use a vector method to find the ratio $BX : XY$

(4)

(ii) Hence find, in its simplest form, the ratio

area of triangle CXY : area of trapezium $ABCD$

(4)

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16. Question 9 · part c.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

17. Question 8 · part d · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part d only.

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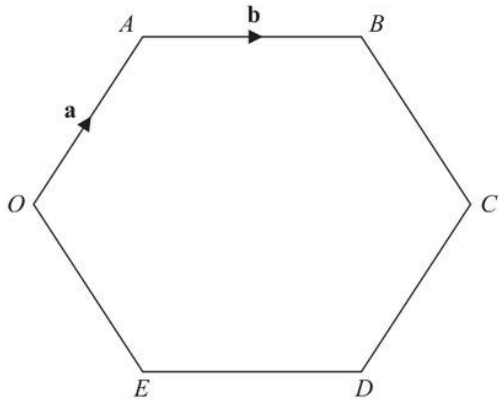


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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17. Question 8 · part d · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

18. Question 8 · part c · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

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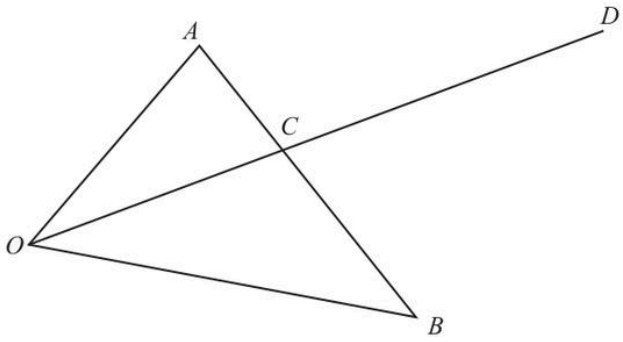


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$ **(3)**

(b) Find the area of triangle AOB **(4)**

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$ **(5)**

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18. Question 8 · part c · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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18. Question 8 · part c · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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18. Question 8 · part c · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 12 marks)

18. Question 8 · part c · 5 marks · Working and answer

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19. Question 11 · part b · 9 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

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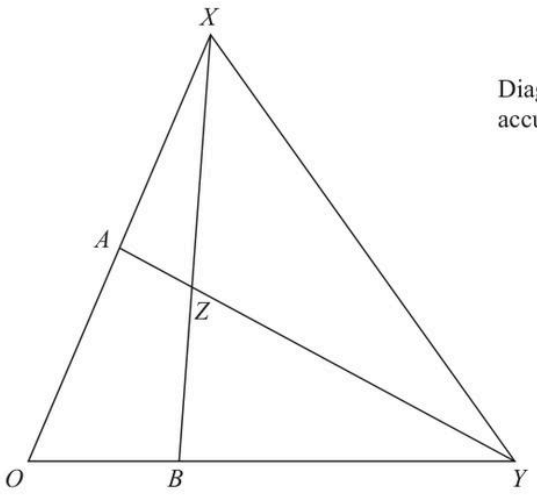


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangle OXY

$\overrightarrow{OX} = 2\mathbf{a}$ and $\overrightarrow{OY} = 3\mathbf{b}$

A is the midpoint of OX and B is the point on OY such that $OB : BY = 1 : 2$
The lines XB and AY intersect at Z.

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

(b) Using a vector method, find $\overrightarrow{OZ}$ as a simplified expression in terms of **a** and **b**

(9)

The point M on XY is such that O, Z and M are collinear.

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(3)

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P 6 2 2 8 3 A 0 3 4 4 0

19. Question 11 · part b · 9 marks · Working and answer

Working and answer

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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

11

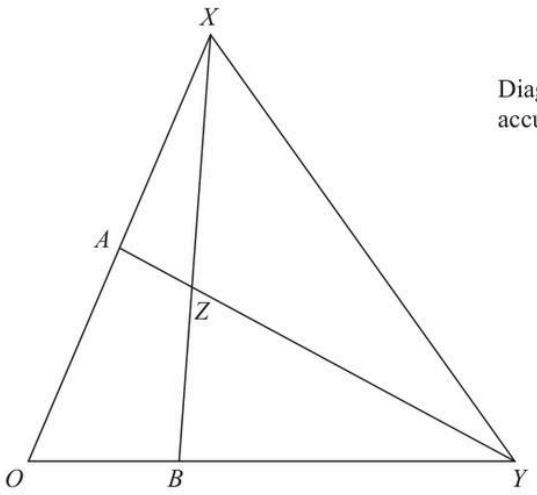


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangle OXY

$$\overrightarrow{OX} = 2\mathbf{a} \text{ and } \overrightarrow{OY} = 3\mathbf{b}$$

A is the midpoint of OX and B is the point on OY such that $OB : BY = 1 : 2$
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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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20. Question 11 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 13 marks)

20. Question 11 · part c · 3 marks · Working and answer

Working and answer

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